

Comparative DCF: Incremental GDP Impact

AI vs. Trump 2.0 vs. US-Iran War (illustrative, order-of-magnitude)

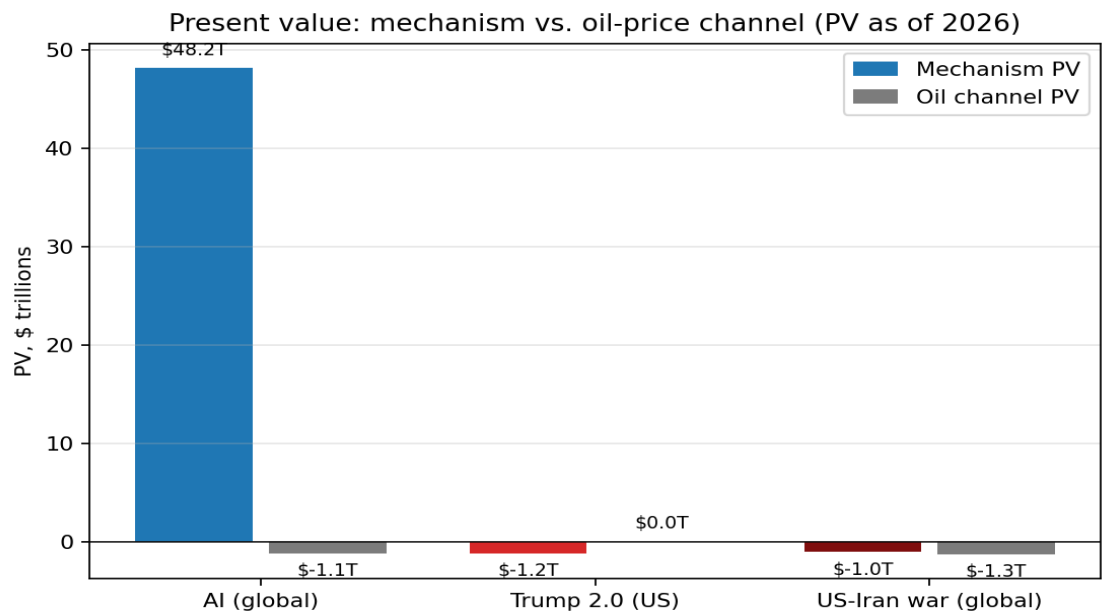
Author: Claude (Opus 4.7) Date: 2026-05-03 Base year: 2026 (year 1 = 2026; PV as of start of 2026) Units: real USD trillions, ex-inflation discount rates Channels: mechanism + oil-price

Executive summary

Under central-case assumptions, with year 1 = 2026 and PV measured as of start-of-2026: AI's incremental GDP has a present value of **\$47.10T** (global, 30-year explicit + Gordon terminal, discount rate 6%; oil channel contributes -\$1.13T). Trump 2.0's incremental US GDP has a present value of **-\$1.13T** (20-year horizon, discount rate 5%; oil channel contributes \$0.03T). A US-Iran kinetic war has a present value of **-\$2.28T** in incremental global GDP (10-year horizon, discount rate 5%; oil channel contributes -\$1.25T and is the dominant transmission mechanism). Conditional on the war occurring — not probability-weighted.

The order of magnitude is the headline: AI's PV is roughly **42x** the absolute size of Trump 2.0's net GDP impact and **21x** the absolute size of an Iran war shock. Even after loosening AI assumptions toward Acemoglu-style skepticism, AI dominates by at least one order of magnitude in our central case — because it is a long-duration productivity shock compounded over decades, while political and geopolitical shocks act on shorter horizons with bounded amplitude.

Scenario	Horizon	Discount	Mechanism PV	Oil PV	Terminal PV	Total PV
AI (global)	30 y	6%	\$25.02T	-\$1.13T	\$23.22T	\$47.10T
Trump 2.0 (US)	20 y	5%	-\$1.16T	\$0.03T	\$0.00T	-\$1.13T
US-Iran war (global)	10 y	5%	-\$1.03T	-\$1.25T	\$0.00T	-\$2.28T



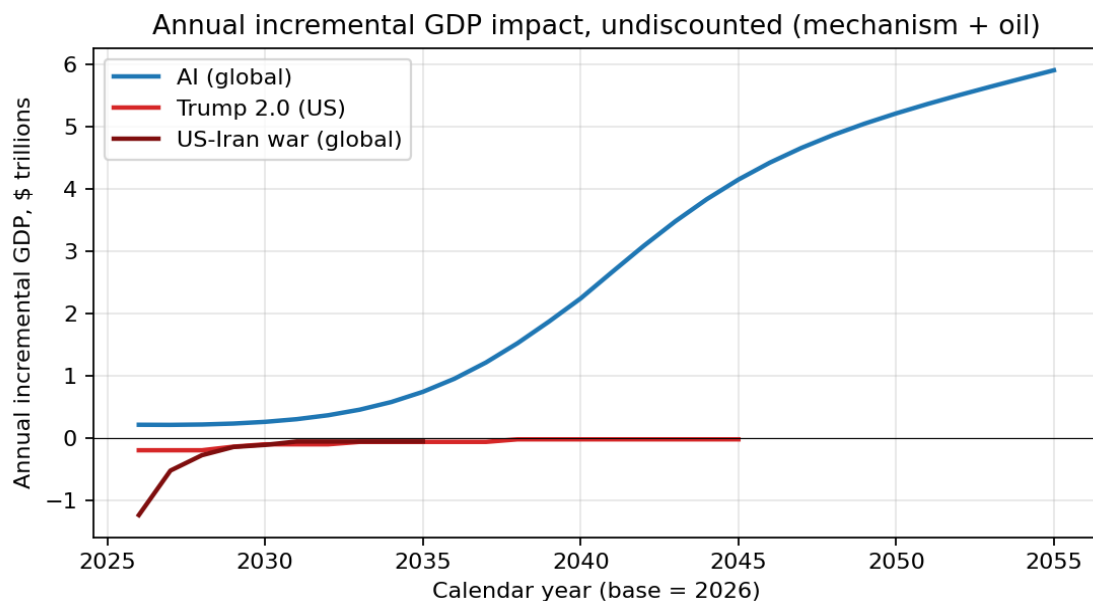
Methodology

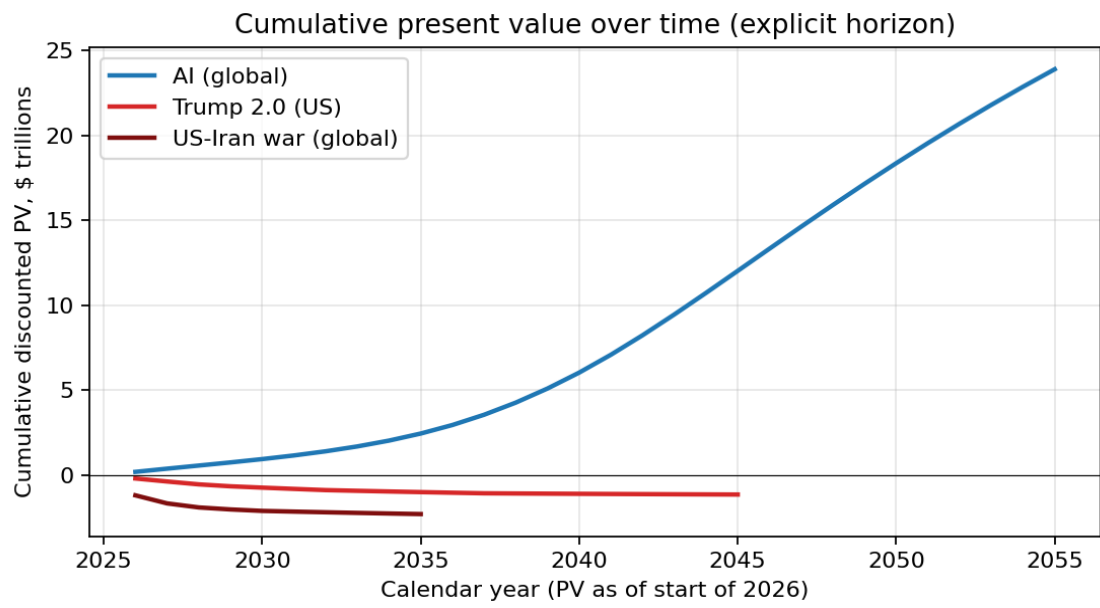
I model each scenario as an annual stream of incremental GDP relative to a no-event counterfactual, starting in 2026, then discount to a present value as of start-of-2026 using a real (ex-inflation) discount rate. $PV = \sum \Delta GDP_t / (1+r)^t$, with a Gordon-growth terminal value where economic fundamentals justify it (only AI). For the political and war scenarios, the impact decays inside the explicit horizon, so no terminal value is added.

Two channels are aggregated for each scenario: (a) the *core mechanism* (productivity, policy, conflict) and (b) an *oil-price channel* that converts a \$/bbl deviation vs a \$75 baseline into GDP impact via an oil-to-GDP elasticity (0.15% global GDP per \$10/bbl sustained for global scenarios; 0.05% US GDP per \$10/bbl for the now oil-balanced US). This makes it possible to see, for each scenario, how much of the headline PV is driven by oil-price assumptions specifically.

Three deliberate choices shape the comparison: (1) AI is global and very long-horizon, while Trump 2.0 is mostly US and 4-20y, while the Iran war shock is global but front-loaded and short. (2) The discount rate is held between 5-6% real to keep magnitudes comparable, then varied in sensitivity tables. (3) War PV is conditional on occurrence, not probability-weighted; if you assign a probability p , multiply by p for an expected value.

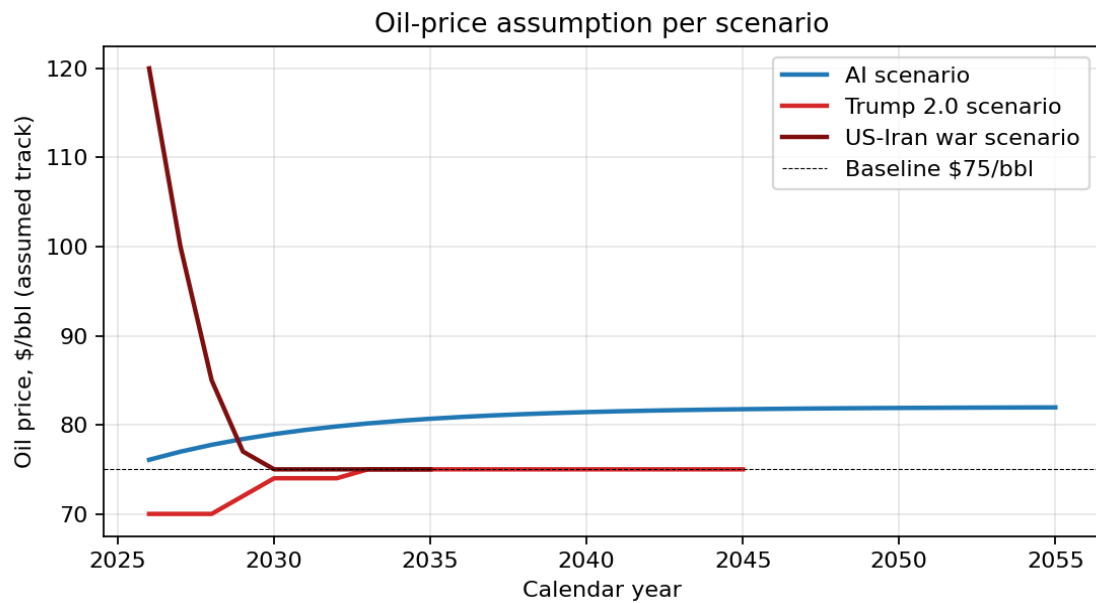
Charts





Oil-price assumption per scenario

Three different oil tracks. The Iran-war track is a sharp spike that dominates that scenario's PV. The AI track is a slow, persistent rise from datacenter energy demand. The Trump 2.0 track is a small drop from US production growth and tariff-driven demand softening, fading after the term.



1) AI — global incremental GDP

Assumptions:

Global GDP base (2026)	\$110T
Adoption profile	Logistic ramp; \$0.2T (2026) -> \$4.5T (2040)
Post-peak growth	+2%/yr through year 30 (2055)
Explicit horizon	30 years (2026-2055)
Terminal growth	1.5% real
Real discount rate	6%
Oil channel	Datacenter electricity demand pushes WTI +\$7/bbl by 2035, persists; -0.10% global GDP drag at
Anchors	Goldman Sachs 2023 (~7% global GDP / 10y); McKinsey 2023 (\$13-25T/yr by 2040); PwC 2017

Result: Explicit-horizon PV \$23.88T + terminal PV \$23.22T = **total PV \$47.10T**. Of which the oil channel contributes -\$1.13T (a small drag from datacenter-driven oil prices). The terminal value is large because a permanent productivity uplift compounds forever; even at 6% real discounting, a \$7-8T perpetual annual delta is worth on the order of \$100T+ in present value terms.

Sensitivity (total PV, \$T):

Peak \$T/yr by 2040 \ Discount	4%	6%	8%	10%
\$2.0T	\$48.04T	\$21.08T	\$11.79T	\$7.45T
\$4.5T	\$109.06T	\$47.10T	\$25.83T	\$15.94T
\$8.0T	\$194.50T	\$83.52T	\$45.48T	\$27.82T
\$12.0T	\$292.15T	\$125.15T	\$67.94T	\$41.40T

Caveats: (i) Acemoglu (MIT, 2024) argues only 0.5% cumulative TFP gain over 10 years — an order of magnitude lower than the central case. (ii) This ignores distributional, employment, and existential-risk side effects, which are not GDP-additive but materially affect welfare. (iii) Capex needed to realize the gain (datacenters, energy, chips) is subtracted from output but not netted here.

2) Trump 2.0 — US incremental GDP

Assumptions (level effects on US GDP):

TCJA extension / corporate tax	+0.4% of US GDP (Tax Foundation)
Tariffs (broad + targeted)	-0.8% of US GDP (Penn Wharton, Yale Budget Lab)
Immigration restriction	-0.4% of US GDP (CBO labor force)
Deregulation	+0.3% of US GDP
Deficit / higher real rates	-0.2% of US GDP
Net peak (mechanism)	-0.7% of US GDP
US GDP base (2026)	\$29T
Profile	Full peak y1-3 (2026-28), handoff y4 (2029), decay through 2045
Real discount rate	5%
Oil channel	WTI -\$5/bbl during term (drill-baby-drill + tariff demand softening); fades by 2033
Oil GDP elasticity	0.05% US GDP per \$10/bbl (US is roughly oil-balanced)

Result: Total PV **-\$1.13T**, of which the mechanism channel is -\$1.16T and the oil channel is \$0.03T (cheaper oil from US production growth is a small positive offset). The sign of the mechanism is negative in the central case but the magnitude is small relative to AI: a 0.7% level shock on a \$29T economy that decays over 20 years is ~\$2-2.5T undiscounted, ~\$1-1.5T discounted. Reasonable bull cases (deregulation > tariff drag) flip the sign without changing the order of magnitude.

Sensitivity (total PV, \$T):

Peak %GDP \ Discount	3%	5%	7%
-1.5% (bear)	-\$2.72T	-\$2.45T	-\$2.23T
-0.7% (base)	-\$1.25T	-\$1.13T	-\$1.03T
0%	\$0.03T	\$0.03T	\$0.03T
+0.7% (bull)	\$1.31T	\$1.18T	\$1.08T

Caveats: (i) Excludes spillovers to non-US economies (likely negative via tariff retaliation but partially offset by reshoring). (ii) Treats policy as a level shock; if Trump 2.0 alters the long-run global trade architecture (durable bloc-formation), the persistent drag could be larger and longer. (iii) Excludes any AI-policy interaction; deregulation around energy/datacenters could be a meaningful AI tailwind.

3) US-Iran war — global incremental GDP

Assumptions (conditional on conflict, central case):

Oil price track (channel)	\$75 (2026 baseline) -> \$120 Y1 -> \$100 Y2 -> \$85 Y3 -> \$77 Y4 -> baseline
Oil-to-GDP elasticity	0.15% global GDP per \$10/bbl sustained
Strait of Hormuz / shipping (mech)	-0.2% of global GDP, year 1 only
Direct US fiscal cost (mech)	\$300-700B over 5y (mid \$500B), -0.05% global GDP/yr
Financial conditions tightening (mech)	-0.2% of global GDP, year 1 only
Long-tail Mid-East instability (mech)	-0.05% of global GDP, years 6-10
Global GDP base (2026)	\$110T
Real discount rate	5%
Probability weighting	NOT applied (conditional PV)

Result: Total PV **-\$2.28T**, of which the oil channel alone is **-\$1.25T** (~55% of the total). Non-oil mechanism channels (Hormuz shipping, fiscal cost, financial tightening, instability) contribute **-\$1.03T**. Front-loaded: ~70% of the loss falls in years 1-3 (2026-2028). If you assign a 20% probability of an actual kinetic war over the next 4 years, the expected-value PV is roughly **-\$0.46T**; at 5%, **-\$0.11T**.

Sensitivity (total PV, \$T) — conditional:

Severity \ Discount	3%	5%	7%
0.5x (limited strikes)	-\$1.19T	-\$1.14T	-\$1.09T
1.0x (base)	-\$2.38T	-\$2.28T	-\$2.19T
1.5x (extended air war)	-\$3.58T	-\$3.42T	-\$3.28T
2.5x (Hormuz closure)	-\$5.96T	-\$5.70T	-\$5.46T

Caveats: (i) Hormuz closure is a fat-tail scenario: ~20% of global oil and ~30% of LNG transit. A 3-month closure could push the shock to multiples of the central case. (ii) Excludes nuclear escalation (non-modelable in DCF terms). (iii) Excludes regional spillovers to Israel-Lebanon, Yemen-Saudi shipping. (iv) Defense Keynesian-stimulus offset is small (~0.05% US GDP) and embedded in the fiscal-cost line.

Comparison & interpretation

Three observations matter more than any single number:

1. Duration dominates magnitude. AI's PV is ~42x Trump 2.0's not because the annual flow is ~42x larger — at peak it's only ~22x — but because AI compounds over a 30-year explicit horizon plus a perpetual terminal value, while Trump 2.0 decays inside 20 years and the war shock decays inside 10. DCF rewards persistence heavily.

2. Sign is contested only for Trump 2.0. AI's central PV is robustly positive across plausible parameterizations (it stays positive even if you halve the peak and double the discount rate). The Iran-war PV is robustly negative. Trump 2.0 is the only scenario where reasonable analysts disagree on the sign, because the policy bundle nets several large opposing forces.

3. The framing arbitrages probability. AI is treated here as a near-certainty (the technology exists; deployment is the question). The war is conditional on occurrence. If you probability-weight (e.g., 20% chance of war) and risk-weight (apply a higher discount rate to AI for deployment risk), the gap narrows but does not close. AI remains the dominant economic force in every scenario where it is not actively suppressed.

What this analysis is NOT

Not a welfare analysis: GDP omits distribution, leisure, environmental stocks, and existential risk. Not a forecast: assumptions are deliberate central cases anchored to published estimates. Not investment advice. Not probability-weighted for the war scenario. The headline ratios are stable across reasonable assumption changes; the absolute numbers should be read as orders of magnitude (one significant figure).